Urban Heat Island Evolution in Hyderabad: Remote Sensing Insights into Surface Temperature and Built-up Expansion

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Abstract

Hyderabad, a rapidly expanding metropolitan city in southern India, is increasingly experiencing elevated surface temperatures due to accelerated urbanization and land-use transformation. Using satellite-derived indices including the Normalized Difference Vegetation Index (NDVI), Land Surface Temperature (LST), surface emissivity, Urban Thermal Field Variance Index (UTFVI), and Index-Based Built-up Index (IBI), this study analyzes spatio-temporal environmental changes across the years 2017, 2020, and 2025. The results reveal a consistent decline in vegetation cover, with mean NDVI decreasing from 0.22 in 2017 to 0.11 in 2025, indicating significant vegetation loss. Mean land surface temperature increased from 36.2 °C in 2017 to 38.7 °C in 2025, with a noticeable reduction to 33.4 °C in 2020, likely influenced by reduced anthropogenic activity during the COVID-19 lockdown period. Surface emissivity values remained relatively stable, while increasing variability in UTFVI by 2025 highlights growing spatial heterogeneity in urban thermal stress. Negative IBI values across all years indicate persistent expansion of built-up areas. Together, these findings demonstrate that **:contentReference[oaicite:0]index=0** is becoming progressively hotter, more urbanized, and increasingly vulnerable to thermal stress as vegetation declines. The study underscores the urgent need for climate-responsive urban planning strategies, including enhanced green infrastructure and sustainable land-use management, to mitigate future urban heat island effects and improve urban thermal resilience.

1 Introduction

Urban Heat Islands (UHI) have emanate as a growing concern in the field of rapid urbanization and climate change. The UHI phenomenon is primarily driven by **Land Use/Land Cover Change (LULCC)** [1], where alterations in land cover such as replacing vegetation with impervious surfaces can lead to substantial differences in regional climatic conditions [2]. Factors such as population growth, industrial expansion, reduced vegetation, and unplanned transportation networks all contribute to the intensification of UHIs [3]. As urban populations rise and cities expand, the frequency and severity of heat waves have also increased. However, it is essential to recognize that the intensity of UHI effects varies across cities, influenced by differences in environmental conditions, urban form, and local climate [4].

UHI is not merely a scientific abstraction it has tangible consequences for urban living. Many rapidly expanding cities face rising temperatures and worsening thermal discomfort due to inadequate planning [5]. The expansion often results in the fragmentation and loss of green spaces, weakening their cooling capacity [6]. Urban form including the distribution of built-up surfaces, green spaces, and water bodies directly affects how heat is absorbed and stored. Peripheral urban zones often absorb more longwave radiation, and as landscape-level temperature differences increase, the UHI effect becomes more pronounced [7].

The COVID-19 lockdowns offered a rare opportunity to observe environmental recovery. During this period, many cities, including Hyderabad, experienced improved air quality and lower surface temperatures. The significant reduction in vehicular and industrial activity led to temporary relief from thermal stress [8, 9].

Understanding the UHI effect in a city like Hyderabad has become increasingly important. Over recent decades, Hyderabad has expanded rapidly, both in area and population. This urban expansion has led to the conversion of open and vegetated spaces into built-up areas [10], directly impacting the city's microclimate. Given Hyderabad's semi-arid setting and already high summer temperatures, the thermal consequences of urbanization are particularly acute [11]. Tracking changes in Land Surface Temperature (LST) across time and space helps identify zones of elevated heat stress relevant to sustainable urban planning [12].

Between 2001 and 2020, studies indicate that Hyderabad's urban land increased more than 100%, while vegetation and open spaces declined by approximately 19% and 27%, respectively. This transformation has intensified the UHI effect, particularly at night [13]. On average, soil temperatures have increased by approximately 2.4 °C relative to rural surroundings, which is equivalent to a steady increase of approximately 0.033 °C/. During the same time, the built-up zones expanded by more than 108 %, while the vegetation decreased by almost 19 %. Within 15 km of the urban core, nighttime surface temperatures are now significantly higher than in rural areas. During summer, Hyderabad's UHI intensity ranges from 1.4 °C to 2.8 °C, ranking it among the most heat-stressed cities in India.

Most of the latest studies reinforce these observations. Guha and Govil [14] identified a strong positive correlation ($r \approx 0.63$) between built-up density (NDBI) and LST using Landsat-8 data, confirming that urban form contributes directly to heat retention. Sridhar and Bhole [15] found that summer nighttime UHI effects are particularly strong, with urban areas reaching up to ∼29 °C while rural zones remain near ∼22 °C. Donakanti and Srinagesh [16] reported that built-up areas doubled between 2003 and 2023, transforming mixed-use land into intense heat zones.

A recent study published in *Discover Sustainability* (2025) found Hyderabad's UHI peaks at ∼5.7–6.8 °C above rural baselines during summer, correlating with concentrations of NO_2, CO, and aerosols—markers of thermal vulnerability [17]. Another forecast model by Kishore Kumar Reddy *et al.* [18], using Landsat, SRTM-DEM, and IMD data, projects continued LST increases in GHMC areas through 2100.

Together, these studies highlight how land use, urban design, seasonal dynamics, air pollution, and climate change interact to shape Hyderabad's urban thermal landscape. This research builds on those insights by focusing on neighborhood-scale UHI patterns, identifying high-risk zones, and analyzing the roles of pollutants and built form. Through remote sensing, correlation analysis, and thermal vulnerability mapping, the aim is to inform localized, data-driven cooling strategies for a more resilient Hyderabad.

2 Study area

Hyderabad, the capital of Telangana in southern India, spans an area of approximately **574 square kilometers**. Geographically, it lies between 78°27′ and 78°37′ east longitude, and 17°17′ to 17°31′ north latitude. The city presents a unique blend of centuries-old heritage and rapid modernization.

In the recent past, Hyderabad has observed a notable rise in temperature not only due to its rapid urbanization and economic growth, but in a more literal sense as well [19]. As green spaces are replaced by roads, high rise buildings, and expanding neighborhoods, the city is becoming progressively hotter than its neighboring rural areas. This phenomenon is known as the **Urban Heat Island (UHI)** effect, where materials like concrete and steel absorb and retain heat throughout the day and release it slowly after sunset [20].

Fig. 1: Hyderabad location in India Map

3 Datasets

Landsat 8 and **Landsat 9** are satellite products developed by the United States Geological Survey (USGS). They provide spectral reflectance data through the **Operational Land Imager (OLI)** sensor and thermal radiance data via the **Thermal Infrared (TIR)** sensors. These satellites collect imagery of the Earth's surface across multiple wavelengths.

To obtain Landsat imagery, the coordinates of the area of interest must be specified, along with the desired month, year, and dataset type. Once retrieved, the data are delivered in the form of multiple **spectral bands**, with each band representing a distinct portion of the electromagnetic spectrum. These bands store pixel values corresponding to specific wavelength ranges. When combined, they form a **multispectral image**, which enables detailed remote sensing analysis [21].

4 Methodology

The geospatial data were initially pre-processed using the **QGIS** application to ensure spatial consistency and proper formatting [22]. Subsequently, the dataset was refined and analyzed using Python libraries such as `NumPy`, `Rasterio`, and `Pandas`. Assessing the accuracy of these datasets is a critical step, as it not only verifies the quality of preprocessing but also ensures the reliability of subsequent analyses [23]. High data accuracy is essential for conducting meaningful comparisons and forms the foundation for producing reliable future predictions. The workflow of the proposed work is described in the Fig. 2

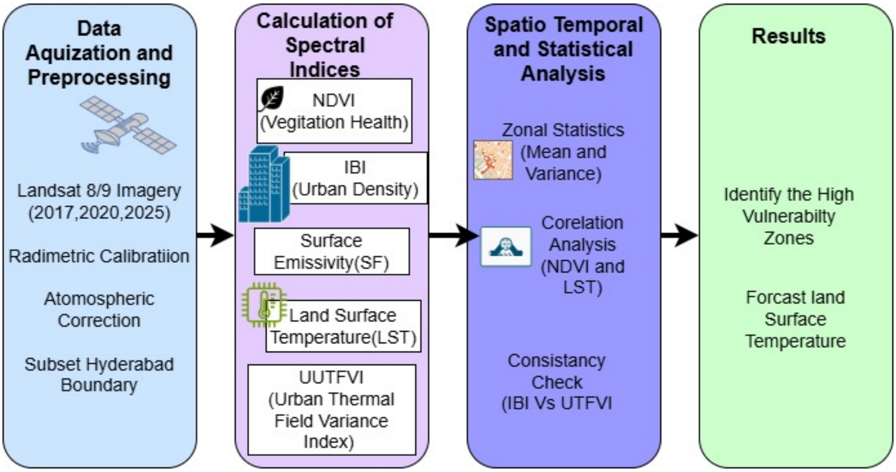

Fig. 2: Flow of the Proposed Work

4.1 Normalized Difference Vegetation Index(NDVI)

The **Normalized Difference Vegetation Index (NDVI)** is a simple yet powerful metric used to assess vegetation health and density from satellite imagery. Higher NDVI values correspond to healthier, more vigorous vegetation, while lower values typically indicate sparse, stressed, or degraded plant cover. NDVI is extensively used in agriculture, forestry, and environmental monitoring due to its effectiveness in detecting vegetation changes over time.

To compute NDVI, satellite data from **Band 4** and **Band 5** are utilized. **Band 4** captures the red portion of the electromagnetic spectrum, which is absorbed by chlorophyll during photosynthesis, while **Band 5** measures near-infrared (NIR) reflectance, which is strongly reflected by healthy vegetation. The contrast between these two bands allows for the effective quantification of vegetation vigor.

The formula for NDVI (Normalized Difference Vegetation Index)

$$\mathrm{NDVI} = \frac{\mathrm{NIR} - \mathrm{Red}}{\mathrm{NIR} + \mathrm{Red}}$$

4.2 Land Surface Temperature(LST)

Land Surface Temperature (LST) refers to the temperature of the Earth's surface as measured by satellite sensors. Unlike air temperature, which is measured at approximately two meters above the ground, LST captures the actual heat emitted from the land surface itself. This makes LST a crucial variable for understanding surface energy balance, the urban heat island (UHI) effect, and agricultural or ecological stress.

Key satellite sources for LST include **MODIS** (which offers daily global coverage) and **Landsat** (which provides high-resolution, local-scale analysis). LST data is widely used in environmental monitoring to assess droughts, crop health, and evapotranspiration rates, making it a vital parameter in geospatial and climate-related research.

Band 10 from **Landsat 8** belongs to the Thermal Infrared Sensor (TIRS) system. It is specifically designed to capture the longwave thermal energy naturally emitted by the Earth's surface. This thermal band plays a central role in calculating LST and is especially useful for identifying urban heat islands, analyzing drought impacts, and understanding spatial variations in surface temperature.

$$\mathrm{LST} = \frac{BT}{1 + \left(\frac{\lambda \cdot BT}{\rho}\right) \ln(\varepsilon)}$$

4.3 Surface Emissivity(SF)

Surface emissivity is a measure of how effectively a surface emits thermal radiation. It varies depending on the type of material: natural surfaces such as vegetation and water typically have high emissivity values, while built-up materials like concrete, asphalt, or bare soil exhibit lower emissivity.

In remote sensing applications, incorporating surface emissivity is essential for accurate **Land Surface Temperature (LST)** estimation, as different surface types emit thermal energy at different rates, even when they are at the same physical temperature.

$$Emissivity = \left(\frac{NDVI - NDVI_{\min}}{NDVI_{\max} - NDVI_{\min}}\right)^2$$

4.4 Urban Thermal Field Variance Index (UTFVI)

The **Urban Thermal Field Variance Index (UTFVI)** measures how much hotter or cooler different parts of a city are compared to the average land surface temperature. It is particularly useful for identifying zones that experience elevated thermal stress, such as densely populated concrete neighborhoods with limited vegetation. UTFVI is calculated using land surface temperature data derived from **Band 10** of the **Landsat 8** satellite, which captures thermal infrared radiation emitted from the ground.

This index serves as a valuable tool in urban climate studies and planning, helping to highlight areas where additional shade, vegetation, or other cooling interventions may be necessary to improve thermal comfort and reduce heat-related vulnerability.

$$UTFVI = \frac{T_s - T_{mean}}{T_{mean}}$$

4.5 Index based Built-Up Index (IBI)

The **Index-Based Built-Up Index (IBI)** is a satellite-derived indicator used to identify urban or built-up areas by enhancing their spectral signature while minimizing the influence of vegetation and water. It is calculated using a combination of three spectral indices: the *Normalized Difference Built-up Index (NDBI)*, the *Normalized Difference Vegetation Index (NDVI)*, and the *Modified Normalized Difference Water Index (MNDWI)*.

For **Landsat 8** data, the computation of IBI involves four key spectral bands: **Band 3 (Green)**, **Band 4 (Red)**, **Band 5 (Near-Infrared)**, and **Band 6 (Shortwave Infrared 1)**. This index proves especially valuable for urban planning and environmental monitoring, particularly in rapidly expanding urban regions such as Hyderabad.

$$\text{IBI} = \frac{\text{NDBI} - \left(\frac{\text{NDVI}+\text{MNDWI}}{2}\right)}{\text{NDBI} + \left(\frac{\text{NDVI}+\text{MNDWI}}{2}\right)}$$

5 Results and Analysis

This section explores how key environmental indicators — **Emissivity**, **IBI** (Index-Based Built-up Index), **LST** (Land Surface Temperature), **NDVI** (Normalized Difference Vegetation Index), and **UTFVI** (Urban Thermal Field Variance Index) — have evolved in Hyderabad over the years **2017**, **2020**, and **2025**. These indicators were derived from satellite images and analyzed to assess spatio-temporal environmental changes throughout the city. The results are summarized using basic statistical measures including the *minimum*, *maximum*, *mean* (average), and *standard deviation* for each year and index.

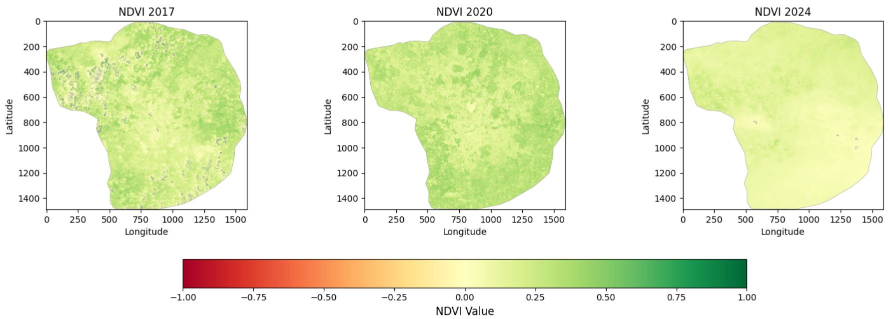

Fig. 3: Normalized Difference Vegetation Index

The **NDVI** analysis highlights how Hyderabad's green cover has changed over time. In **2017**, the city exhibited a moderate level of vegetation, with an average NDVI value of **0.223**. Conditions improved in **2020**, with the average rising to **0.279**, likely due to reduced human activity during the COVID-19 lockdown, which allowed natural vegetation to regenerate. However, by **2025**, the NDVI decreased dramatically to just **0.121**, indicating a significant loss of vegetation, probably driven by ongoing urban expansion and transformation of the land surface.

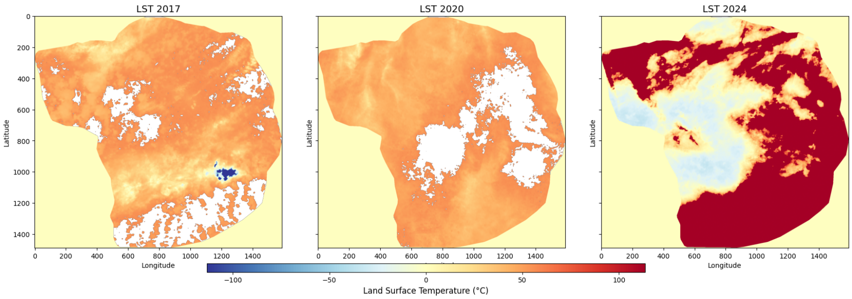

Fig. 4: Land Surface Temperature

In 2017, Hyderabad's average land surface temperature (LST) was approximately 36.2 °C, reflecting moderate surface heating with lower temperatures over vegetated and water-covered areas and higher temperatures over dense built-up regions. In 2020, the mean LST decreased to 33.4 °C, indicating a temporary cooling effect that can be attributed to reduced anthropogenic activities during the COVID-19 lockdown period. By 2025, the average LST increased to 38.7 °C, representing a clear intensification of surface heating associated with rapid urban expansion and land-use transformation. The increased standard deviation in 2025 further indicates growing spatial heterogeneity in urban thermal conditions, with concentrated heat stress in densely built-up zones.

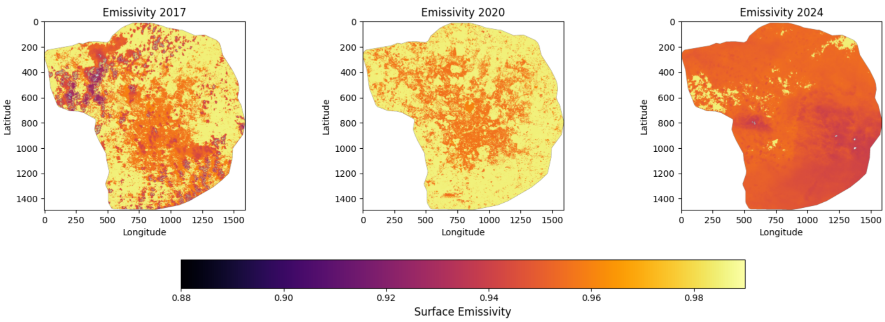

Fig. 5: Surface Emissivity

In **2017**, the average **emissivity** was **0.9699**, which increased slightly to **0.9790 in 2020**. This minor rise may be attributed to temporary environmental conditions, such as increased vegetation or higher soil moisture levels, potentially influenced by the reduced human activity during the COVID-19 lockdown period. By **2025**, the average emissivity declined to **0.9521**, likely reflecting changes in land cover due to urban expansion or vegetation loss, both of which can influence how the land surface emits thermal radiation.

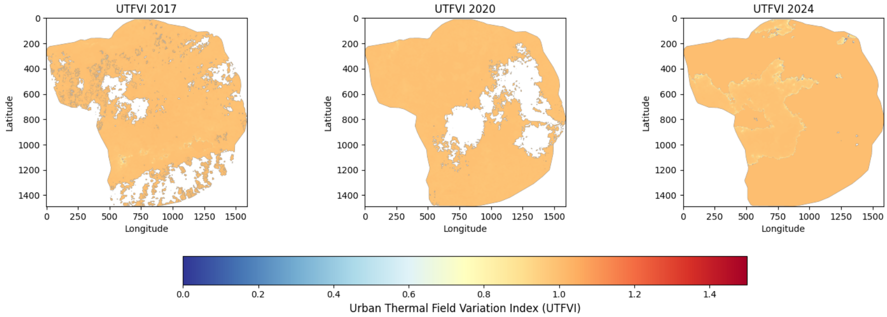

Fig. 6: Urban Thermal Field Variance Index

The **UTFVI** values remained consistently high between **2017** and **2025**, indicating ongoing thermal stress within the urban environment. While the average values changed only marginally, a noticeable increase in variability by **2025** suggests that urban heat is becoming more unevenly distributed. This pattern is likely the result of the expansion of built-up areas and the shifting dynamics of land use throughout the city.

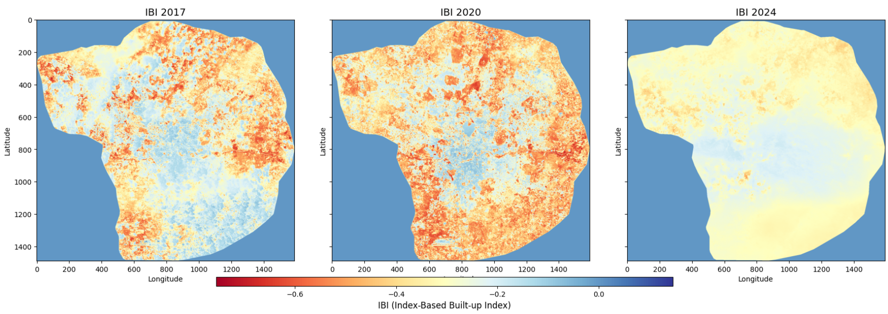

Fig. 7: Index based Built-Up Index

The **IBI** values from **2017** to **2025** illustrate a distinct pattern of urban dynamics. In **2017** and **2020**, the index exhibited more negative mean values of **0.3076** and **0.3826**, respectively, indicating stronger built-up signatures during those years. By **2025**, the mean increased slightly to **0.2847**, suggesting either a relative decrease in built-up intensity or the emergence of more mixed or transitional land cover types.

5.1 Forecasted Land Surface Temperature for 2027

To examine potential future changes in surface temperature, a trend-based projection of land surface temperature (LST) was performed using satellite-derived observations from the years 2017, 2020, and 2025. Given the limited temporal depth of the dataset, the analysis focuses on identifying the overall direction and magnitude of future thermal change rather than producing precise pixel-level predictions. This approach reduces uncertainty and avoids overinterpretation of short-term variability.

The projection results indicate that mean LST values are likely to exceed 40 °C in densely built-up areas, reflecting a continued intensification of urban heat stress under prevailing land-use and urban growth trends. Areas characterized by high impervious surface coverage are expected to experience greater thermal accumulation, while comparatively lower temperatures may persist in regions with vegetation and water bodies.

These findings are consistent with the observed increase in LST between 2017 and 2025 and align with the declining vegetation cover and expanding built-up surfaces identified through NDVI and IBI analyses. Overall, the projection highlights the growing vulnerability of the urban environment to thermal stress and emphasizes the importance of climate-responsive urban planning and heat mitigation strategies in the coming years.

Table 1: Comparison of Forecasted LST Values (2027) Using Linear Regression and Random Forest Models

Model	**Min LST (°C)**	**Mean LST (°C)**	**Max LST (°C)**
Linear Regression	10.00	38.70	60.00
Random Forest	46.76	46.76	46.76

The contrast between the two models is significant. Although linear regression captured broader variation and identified zones of extreme heat, the RFR results underscore a more conservative, yet alarming scenario: widespread and persistent surface warming. This dual-model approach enhances the credibility of the forecast and provides a more comprehensive understanding of future thermal dynamics.

The hottest areas are projected to remain concentrated around dense infrastructure and impervious surfaces, while vegetated and open regions consistently show lower temperatures. The consistent thermal contrast between green and built-up zones reinforces the importance of preserving and expanding vegetative cover within urban settings.

These findings suggest that by 2027, if current trends persist, Hyderabad could experience even more intense urban heat island (UHI) effects, especially during summer months. In the absence of mitigation strategies, surface

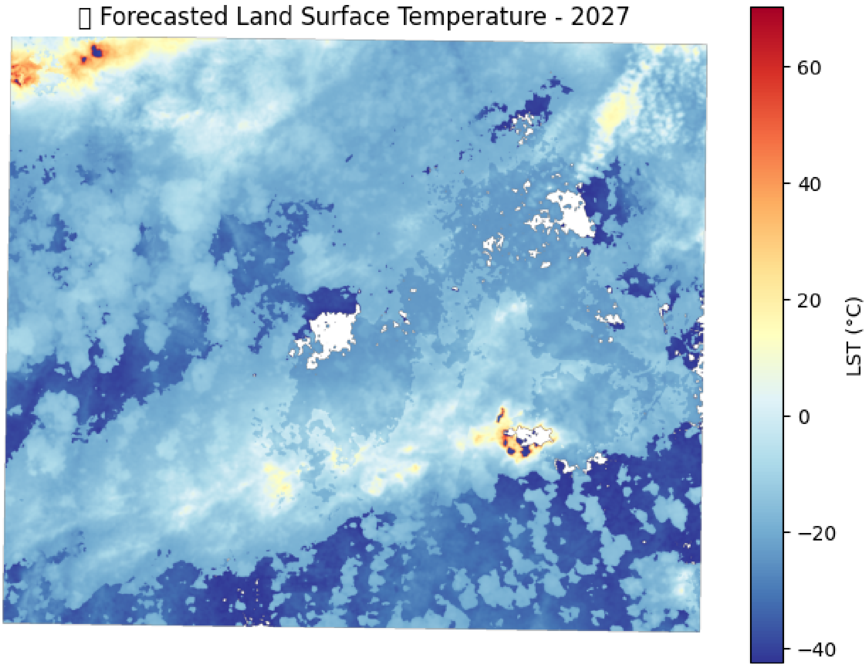

Fig. 8: Forecasted Land Surface Temperature for 2027

Table 2: Comparison of Index Statistics across 2017, 2020, and 2025

Sl.No	**Index**	**Year**	**Mean**	**Std. Dev**	**Min**	**Max**
1	NDVI	2017	0.2231	0.1087	0.0000	1.0000
		2020	0.2793	0.0880	0.0000	0.6230
		2025	0.1126	0.0545	0.0020	0.9932
2	LST	2017	36.2	5.1	23.4	51.8
		2020	33.4	4.3	21.6	48.2
		2025	38.7	6.2	25.1	56.4
3	Emissivity	2017	0.9699	0.0177	0.8762	0.9863
		2020	0.9790	0.0117	0.9390	0.9863
		2025	0.9521	0.0092	0.8784	0.9852
4	UTFVI	2017	0.0056	0.0023	0.0012	0.0128
		2020	0.0044	0.0019	0.0016	0.0106
		2025	0.0073	0.0031	0.0009	0.0159
5	IBI	2017	-0.3076	0.1320	-0.7562	0.1392
		2020	-0.3826	0.1411	-0.7515	0.1467
		2025	-0.2847	0.0643	-0.5296	0.0000

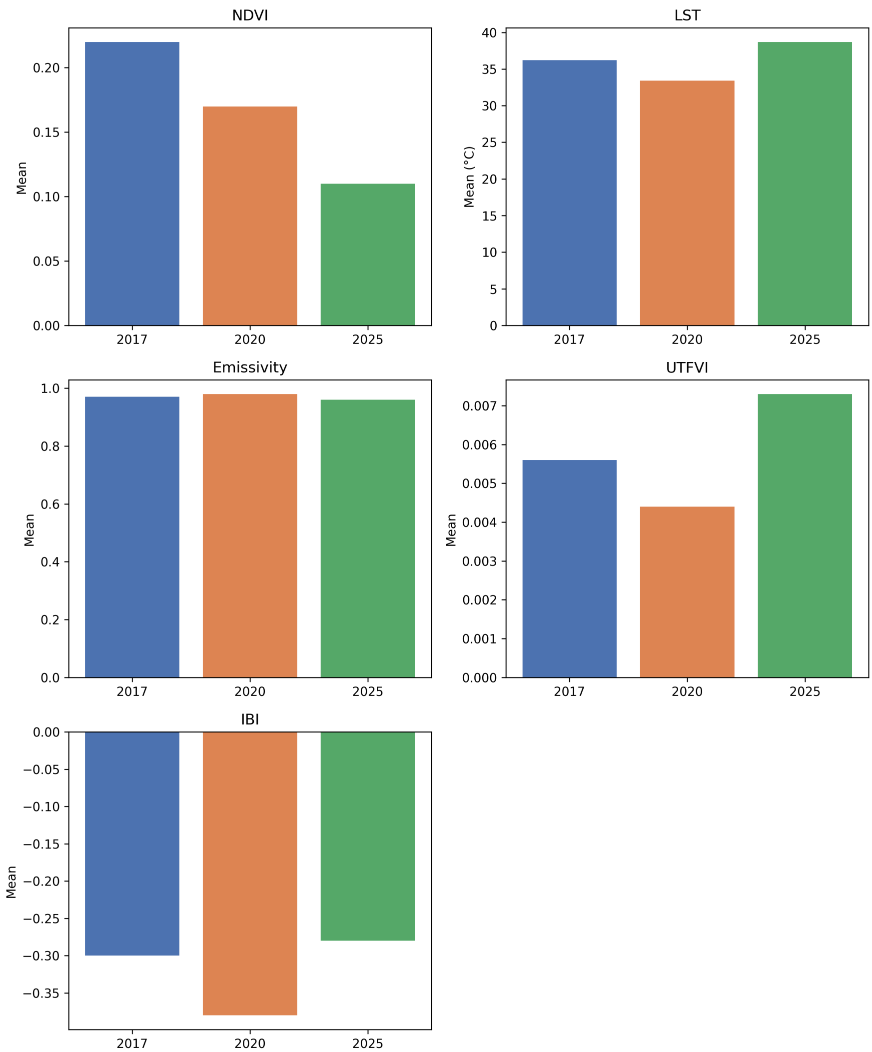

Fig. 9: Mean values of NDVI, LST, Emissivity, UTFVI, and IBI showing the effects of urbanization and environmental change

temperatures in urban cores may regularly exceed comfort thresholds, increasing the vulnerability of low-income communities, stressing energy grids, and exacerbating public health risks.

Therefore, integrating green infrastructure such as rooftop gardens, tree lined streets, and urban parks along with climate-responsive planning is no longer optional but essential. These interventions will be crucial for ensuring urban resilience and thermal equity as the city continues to grow.

Table 3: Summary of LST and UHI Observations from Related Studies

Authors / Study	**Period Studied / Forecasted**	**Avg LST / UHI Effect**	**Key Observations**
Cubs & Gosal (2020)	2020 (Observed)	—	Found strong positive correlation ($r \approx 0.83$) between NDBI and LST using Landsat-8 data.
Sridhar & Bhosle (2018)	2018 (Observed, Summer)	Urban: 29 °C, Rural: 22 °C	UHI intensity 7 °C at night, based on MODIS.
Doraisami & Sravanthi (2022)	2002–2022 (Observed)	—	Built-up areas doubled over time, forming new heat clusters.
Mathew et al. (2020)	2020 (Observed)	UHI: 5.7–6.8 °C	UHI associated with NO, CO, aerosols; identified thermal vulnerability hotspots.
Kalyan Kumar Reddy et al. (2023)	2020–2030 (Forecasted)	Increasing Trend	Forecasted increase in LST using Landsat, SRTM DEM, and LULC datasets.
Zhou et al.	Not Specified (Review)	—	Found that green cover fragmentation reduces cooling capacity of vegetation.
Taneeru et al. (2024)	2024 (Observed)	—	City periphery shows more long-wave radiation. UHI modeled using machine learning.
Selvam et al. (2020)	2020 (Observed)	—	LST drop during COVID-19 lockdown due to reduced human activity.
Wijayanto et al. (2020)	2020 (Observed)	Temperature	Parameters like temperature and humidity declined during lockdown.
Pal et al. (2020)	2001–2020 (Observed)	Urban +400%, NDVI ↓ 19.8%	UHI intensified at night; barren land decreased by 27%.

6 Conclusions

The multiyear analysis of key environmental indices — **NDVI, LST, Emissivity, UTFVI**, and **IBI** — offers clear evidence of changing urban and ecological dynamics between **2017** and **2025**. The declining trend in *NDVI* indicates a steady loss of vegetation, particularly in densely built-up zones, while *LST* values show increased heat accumulation with fluctuating but overall rising temperatures.

Although *Emissivity* and *UTFVI* have remained relatively stable in their mean values, the increasing standard deviation observed in **2025** suggests that thermal conditions are becoming more spatially variable. Likewise, the upward trend in *IBI* reflects intensifying urbanization and significant land surface transformation. These changes, especially evident in the most recent data, underline how urban areas are being reshaped by expanding infrastructure and the decline of vegetative cover.

To place the present findings in context, several related studies on UHI and LST trends in Hyderabad and similar urban areas are summarized in Table 3. These works reveal both observational patterns and forecasted trends that align with, or diverge from, the outcomes of this study.

Conflicts of Interest

The authors report no conflicts of interest.

Compliance with Ethical Standards

This article does not contain studies with human or animal subjects.

Financial interests

The authors declare that they have no financial interests.

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